DATA ANALYTICS LAB SHEET 5

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B.Tech CSE

Experiment No. 5

Interactive Dashboard Development using Tableau / Microsoft Power BI

AIM

To design and develop an interactive business dashboard using a BI tool for analyzing a real-world dataset and presenting key performance indicators (KPIs) to support data-driven decision-making.

PURPOSE

Business Intelligence tools transform raw data into interactive dashboards and reports. Dashboards allow users to filter, explore, and analyze business performance dynamically.

DATASET

Superstore Sales Dataset.

NOTE

This Colab notebook demonstrates the dashboard preparation, KPI calculations, visualizations, filters, and business insights using Python. The same prepared dataset and metrics can be imported into Tableau or Microsoft Power BI to create the final interactive dashboard.

1. Import Required Libraries

Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")

print("Libraries imported successfully.")
```

2. Load the Superstore Dataset

Code:

```
url = "https://raw.githubusercontent.com/leonism/sample-superstore/master/data/superstore.csv"

df = pd.read_csv(url, encoding="latin1")

print("Dataset loaded successfully.")
print("Dataset shape:", df.shape)
display(df.head())
```

```
  Row ID        Order ID  Order Date   Ship Date       Ship Mode Customer ID  \
0      1  CA-2017-152156   11/8/2017  11/11/2017    Second Class    CG-12520
1      2  CA-2017-152156   11/8/2017  11/11/2017    Second Class    CG-12520
2      3  CA-2017-138688   6/12/2017   6/16/2017    Second Class    DV-13045
3      4  US-2016-108966  10/11/2016  10/18/2016  Standard Class    SO-20335
4      5  US-2016-108966  10/11/2016  10/18/2016  Standard Class    SO-20335

     Customer Name    Segment        Country             City  ...  \
0      Claire Gute   Consumer  United States        Henderson  ...
1      Claire Gute   Consumer  United States        Henderson  ...
2  Darrin Van Huff  Corporate  United States      Los Angeles  ...
3   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...
4   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...

  Postal Code  Region       Product ID         Category Sub-Category  \
```

```
0     42420.0   South  FUR-BO-10001798        Furniture    Bookcases
1     42420.0   South  FUR-CH-10000454        Furniture       Chairs
2     90036.0    West  OFF-LA-10000240  Office Supplies       Labels
3     33311.0   South  FUR-TA-10000577        Furniture       Tables
4     33311.0   South  OFF-ST-10000760  Office Supplies      Storage

                                     Product Name     Sales  Quantity  \
0                  Bush Somerset Collection Bookcase  261.9600       2.0
1  Hon Deluxe Fabric Upholstered Stacking Chairs,...  731.9400       3.0
2  Self-Adhesive Address Labels for Typewriters b...   14.6200       2.0
3      Bretford CR4500 Series Slim Rectangular Table  957.5775       5.0
4                    Eldon Fold 'N Roll Cart System   22.3680       2.0

   Discount    Profit
0      0.00   41.9136
1      0.00  219.5820
2      0.00    6.8714
3      0.45 -383.0310
4      0.20    2.5164

[5 rows x 21 columns]
```

3. Verify Dataset Fields and Data Types

Code:

```
print("Column names:")
print(df.columns.tolist())

print("\nData types:")
print(df.dtypes)

print("\nMissing values:")
display(df.isnull().sum().sort_values(ascending=False).head(15))

print("\nDataset summary:")
display(df.describe(include="all").T)
```

```
Postal Code      817
Ship Mode        806
Ship Date        806
Customer ID      806
Order Date       806
Product ID       806
Category         806
Segment          806
Customer Name    806
Country          806
City             806
State            806
Region           806
Sales            806
Quantity         806
dtype: int64
```

```
                count unique              top  freq          mean  \
Row ID          10800  10001              Yes   800           NaN
Order ID        10800   5015   CA-2018-100111    28           NaN
Order Date       9994   1236        9/5/2017    38           NaN
Ship Date        9994   1334       12/16/2016    35           NaN
Ship Mode        9994      4   Standard Class  5968           NaN
Customer ID      9994    793        WB-21850    37           NaN
Customer Name    9994    793    William Brown    37           NaN
Segment          9994      3         Consumer  5191           NaN
Country          9994      1    United States  9994           NaN
City             9994    531    New York City   915           NaN
State            9994     49       California  2001           NaN
Postal Code    9983.0    NaN              NaN   NaN  55245.233297
Region           9994      4             West  3203           NaN
Product ID       9994   1862  OFF-PA-10001970    19           NaN
Category         9994      3  Office Supplies  6026           NaN
Sub-Category     9994     17          Binders  1523           NaN
Product Name     9994   1850  Staple envelope    48           NaN
Sales          9994.0    NaN              NaN   NaN    229.858001
Quantity       9994.0    NaN              NaN   NaN      3.789574
Discount       9994.0    NaN              NaN   NaN      0.156203
Profit         9994.0    NaN              NaN   NaN     28.656896

                       std       min      25%      50%      75%       max
Row ID                 NaN       NaN      NaN      NaN      NaN       NaN
Order ID               NaN       NaN      NaN      NaN      NaN       NaN
Order Date             NaN       NaN      NaN      NaN      NaN       NaN
Ship Date              NaN       NaN      NaN      NaN      NaN       NaN
Ship Mode              NaN       NaN      NaN      NaN      NaN       NaN
Customer ID            NaN       NaN      NaN      NaN      NaN       NaN
Customer Name          NaN       NaN      NaN      NaN      NaN       NaN
Segment                NaN       NaN      NaN      NaN      NaN       NaN
Country                NaN       NaN      NaN      NaN      NaN       NaN
City                   NaN       NaN      NaN      NaN      NaN       NaN
State                  NaN       NaN      NaN      NaN      NaN       NaN
```

```
Postal Code    32038.715955    1040.0  23223.0  57103.0  90008.0   99301.0
Region                  NaN       NaN      NaN      NaN      NaN       NaN
Product ID              NaN       NaN      NaN      NaN      NaN       NaN
Category                NaN       NaN      NaN      NaN      NaN       NaN
Sub-Category            NaN       NaN      NaN      NaN      NaN       NaN
Product Name            NaN       NaN      NaN      NaN      NaN       NaN
Sales            623.245101     0.444    17.28    54.49   209.94  22638.48
Quantity            2.22511       1.0      2.0      3.0      5.0      14.0
Discount           0.206452       0.0      0.0      0.2      0.2       0.8
Profit           234.260108 -6599.978  1.72875   8.6665   29.364  8399.976
```

4. Data Preprocessing

Code:

```
# Convert date fields
df["Order Date"] = pd.to_datetime(df["Order Date"], errors="coerce")
df["Ship Date"] = pd.to_datetime(df["Ship Date"], errors="coerce")

# Create useful dashboard fields
df["Year"] = df["Order Date"].dt.year
df["Month"] = df["Order Date"].dt.month
df["Month Name"] = df["Order Date"].dt.strftime("%b")
df["Order Month"] = df["Order Date"].dt.to_period("M").astype(str)

# Remove rows missing essential analytical values
df = df.dropna(subset=["Sales", "Profit", "Order ID"])

print("Preprocessing completed.")
display(df.head())
```

```
  Row ID        Order ID Order Date  Ship Date       Ship Mode Customer ID  \
0      1  CA-2017-152156 2017-11-08 2017-11-11    Second Class    CG-12520
1      2  CA-2017-152156 2017-11-08 2017-11-11    Second Class    CG-12520
2      3  CA-2017-138688 2017-06-12 2017-06-16    Second Class    DV-13045
3      4  US-2016-108966 2016-10-11 2016-10-18  Standard Class    SO-20335
4      5  US-2016-108966 2016-10-11 2016-10-18  Standard Class    SO-20335

     Customer Name    Segment        Country             City  ...  \
0      Claire Gute   Consumer  United States        Henderson  ...
1      Claire Gute   Consumer  United States        Henderson  ...
2  Darrin Van Huff  Corporate  United States      Los Angeles  ...
3   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...
4   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...

  Sub-Category                                Product Name     Sales  \
0    Bookcases             Bush Somerset Collection Bookcase  261.9600
1       Chairs  Hon Deluxe Fabric Upholstered Stacking Chairs,...  731.9400
2       Labels  Self-Adhesive Address Labels for Typewriters b...   14.6200
3       Tables      Bretford CR4500 Series Slim Rectangular Table  957.5775
4      Storage                  Eldon Fold 'N Roll Cart System   22.3680

  Quantity Discount    Profit    Year  Month  Month Name  Order Month
0      2.0     0.00   41.9136  2017.0   11.0         Nov      2017-11
1      3.0     0.00  219.5820  2017.0   11.0         Nov      2017-11
2      2.0     0.00    6.8714  2017.0    6.0         Jun      2017-06
3      5.0     0.45 -383.0310  2016.0   10.0         Oct      2016-10
4      2.0     0.20    2.5164  2016.0   10.0         Oct      2016-10

[5 rows x 25 columns]
```

5. Key Performance Indicators (KPIs)

Code:

```
total_sales = df["Sales"].sum()
total_profit = df["Profit"].sum()
number_of_orders = df["Order ID"].nunique()
customer_count = df["Customer ID"].nunique()
average_order_value = total_sales / number_of_orders
profit_margin = (total_profit / total_sales) * 100

kpi_table = pd.DataFrame({
    "KPI": [
        "Total Sales",
        "Total Profit",
        "Number of Orders",
        "Customer Count",
        "Average Order Value",
        "Profit Margin (%)"
    ],
    "Value": [
        total_sales,
        total_profit,
        number_of_orders,
        customer_count,
        average_order_value,
        profit_margin
    ]
```

```
})

display(kpi_table)
```

```
                   KPI         Value
0          Total Sales  2.297201e+06
1         Total Profit  2.863970e+05
2     Number of Orders  5.009000e+03
3       Customer Count  7.930000e+02
4  Average Order Value  4.586147e+02
5    Profit Margin (%)  1.246722e+01
```

6. KPI Cards

Code:

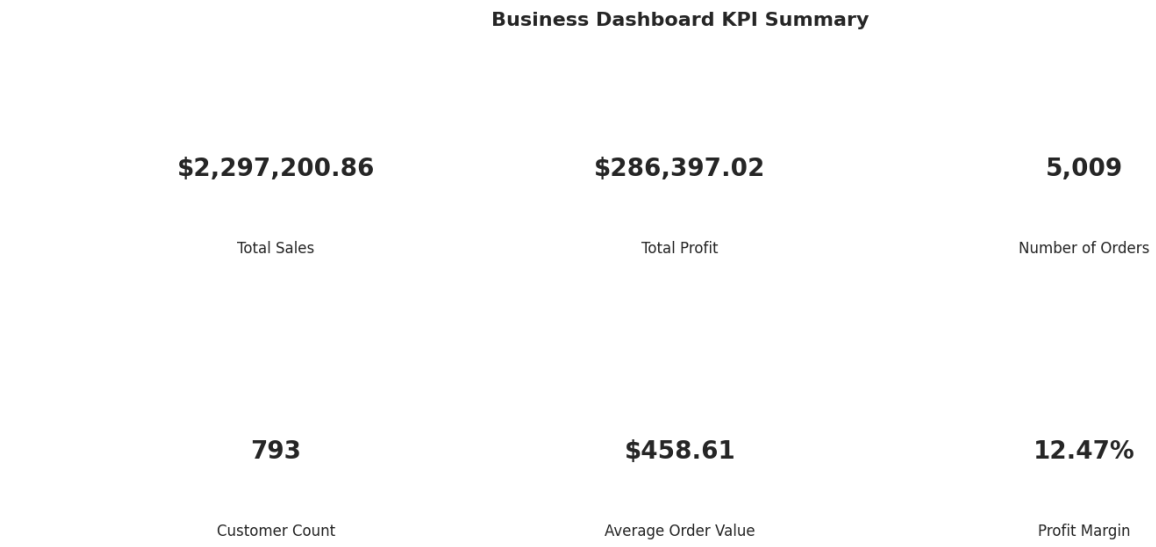
```
fig, axes = plt.subplots(2, 3, figsize=(14, 7))
axes = axes.flatten()

kpis = [
    ("Total Sales", f"${total_sales:,.2f}"),
    ("Total Profit", f"${total_profit:,.2f}"),
    ("Number of Orders", f"{number_of_orders:,}"),
    ("Customer Count", f"{customer_count:,}"),
    ("Average Order Value", f"${average_order_value:,.2f}"),
    ("Profit Margin", f"{profit_margin:.2f}%")
]

for ax, (title, value) in zip(axes, kpis):
    ax.text(0.5, 0.55, value, ha="center", va="center", fontsize=20, fontweight="bold")
    ax.text(0.5, 0.25, title, ha="center", va="center", fontsize=12)
    ax.set_axis_off()

plt.suptitle("Business Dashboard KPI Summary", fontsize=16, fontweight="bold")
plt.tight_layout()
plt.show()
```

```
<Figure size 1400x700 with 6 Axes>
```



7. Sales by Region

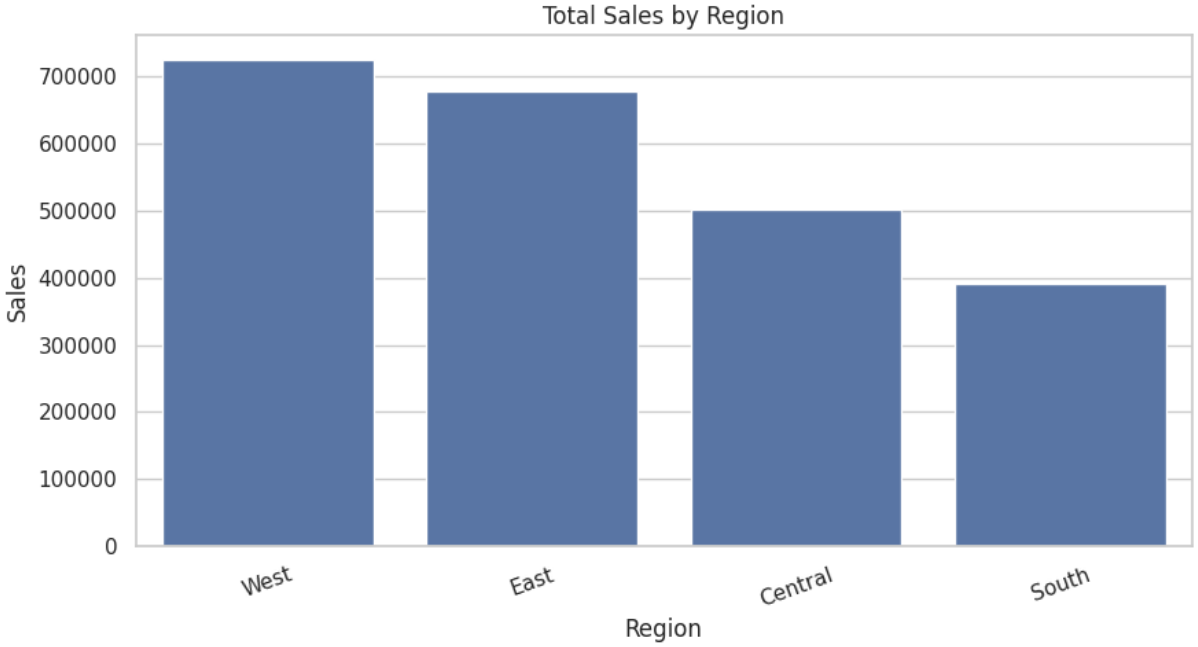
Code:

```
region_sales = df.groupby("Region")["Sales"].sum().sort_values(ascending=False)

plt.figure(figsize=(9, 5))
sns.barplot(x=region_sales.index, y=region_sales.values)
plt.title("Total Sales by Region")
plt.xlabel("Region")
plt.ylabel("Sales")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()

display(region_sales.reset_index(name="Sales"))
```

```
<Figure size 900x500 with 1 Axes>
```




```
    Region        Sales
0     West  725457.8245
1     East  678781.2400
2  Central  501239.8908
3    South  391721.9050
```

8. Sales and Profit by Category

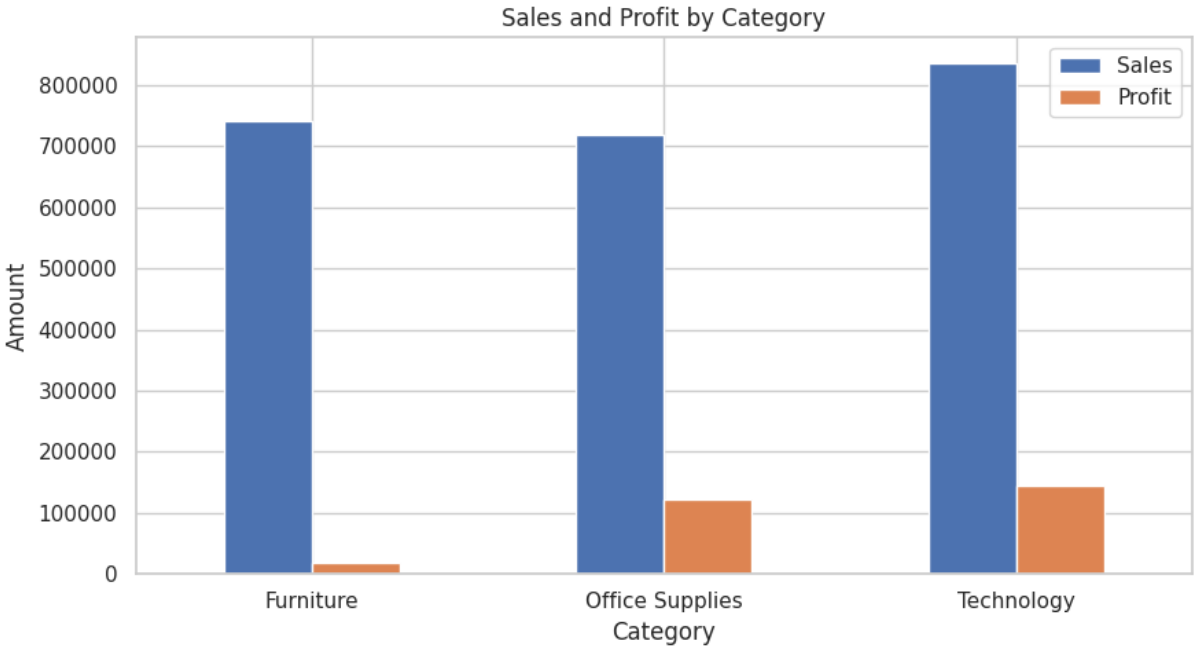
Code:

```
category_summary = df.groupby("Category")[["Sales", "Profit"]].sum()

category_summary.plot(kind="bar", figsize=(9, 5))
plt.title("Sales and Profit by Category")
plt.xlabel("Category")
plt.ylabel("Amount")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()

display(category_summary)
```

```
<Figure size 900x500 with 1 Axes>
```




```
                    Sales       Profit
Category
Furniture        741999.7953   18451.2728
Office Supplies  719047.0320  122490.8008
Technology       836154.0330  145454.9481
```

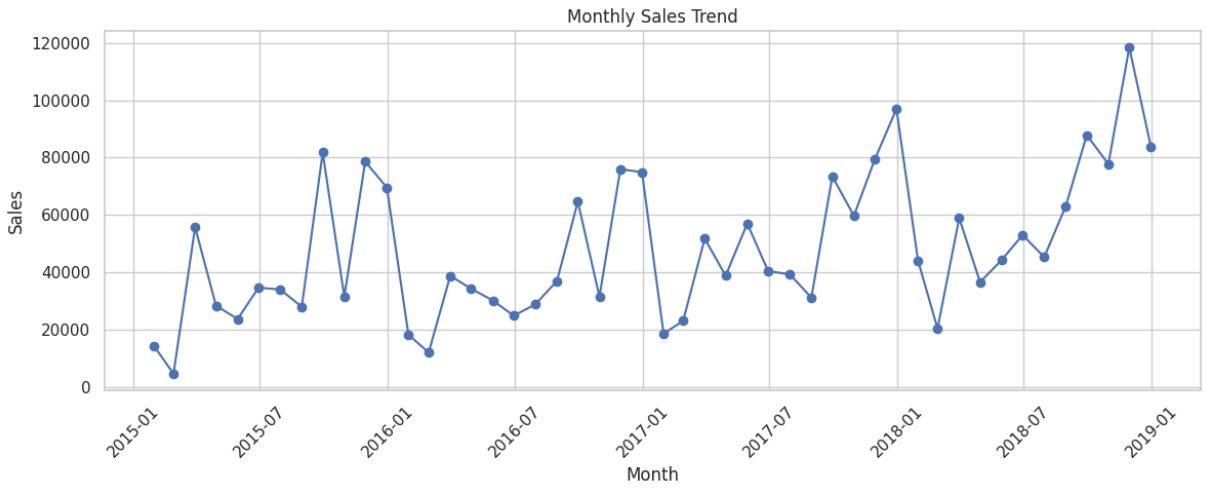
9. Sales Trend Over Time

Code:

```
monthly_sales = (
    df.dropna(subset=["Order Date"])
      .set_index("Order Date")["Sales"]
      .resample("ME")
      .sum()
)

plt.figure(figsize=(12, 5))
plt.plot(monthly_sales.index, monthly_sales.values, marker="o")
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

```
<Figure size 1200x500 with 1 Axes>
```



10. Sales by Customer Segment

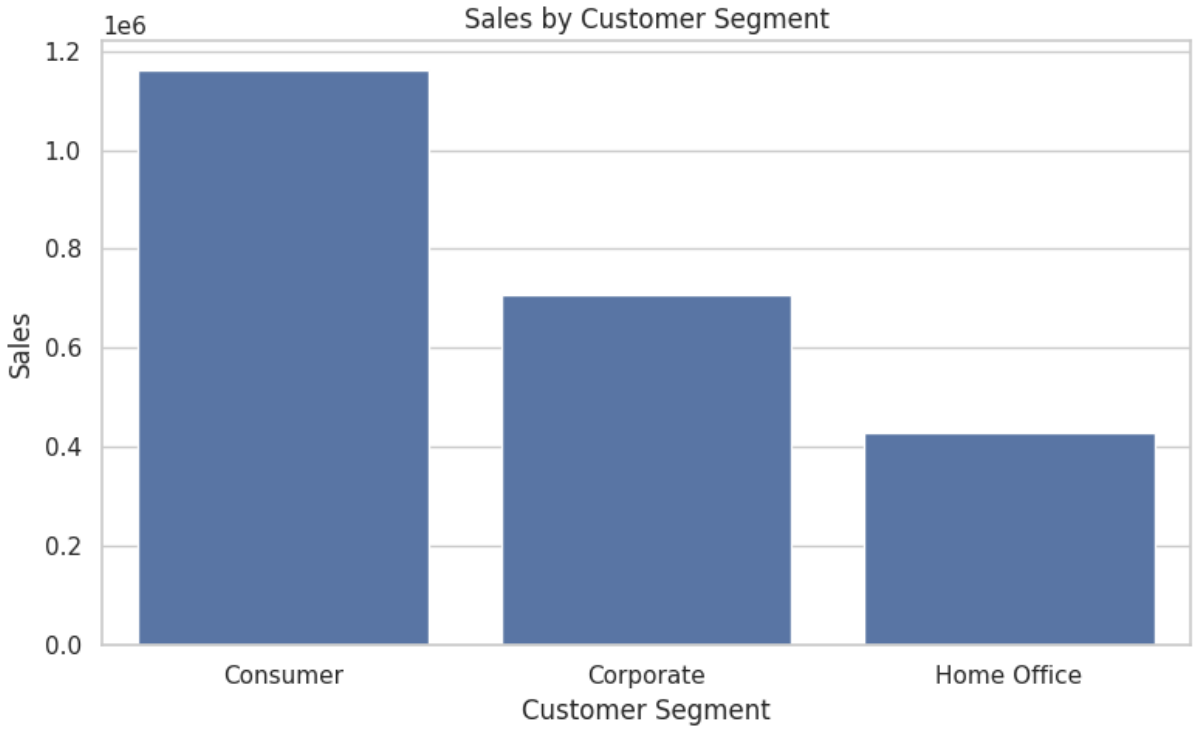
Code:

```
segment_sales = df.groupby("Segment")["Sales"].sum().sort_values(ascending=False)

plt.figure(figsize=(8, 5))
sns.barplot(x=segment_sales.index, y=segment_sales.values)
plt.title("Sales by Customer Segment")
plt.xlabel("Customer Segment")
plt.ylabel("Sales")
plt.tight_layout()
plt.show()

display(segment_sales.reset_index(name="Sales"))
```

```
<Figure size 800x500 with 1 Axes>
```




```
       Segment         Sales
0     Consumer  1.161401e+06
1    Corporate  7.061464e+05
2  Home Office  4.296531e+05
```

11. Top 10 Products by Sales

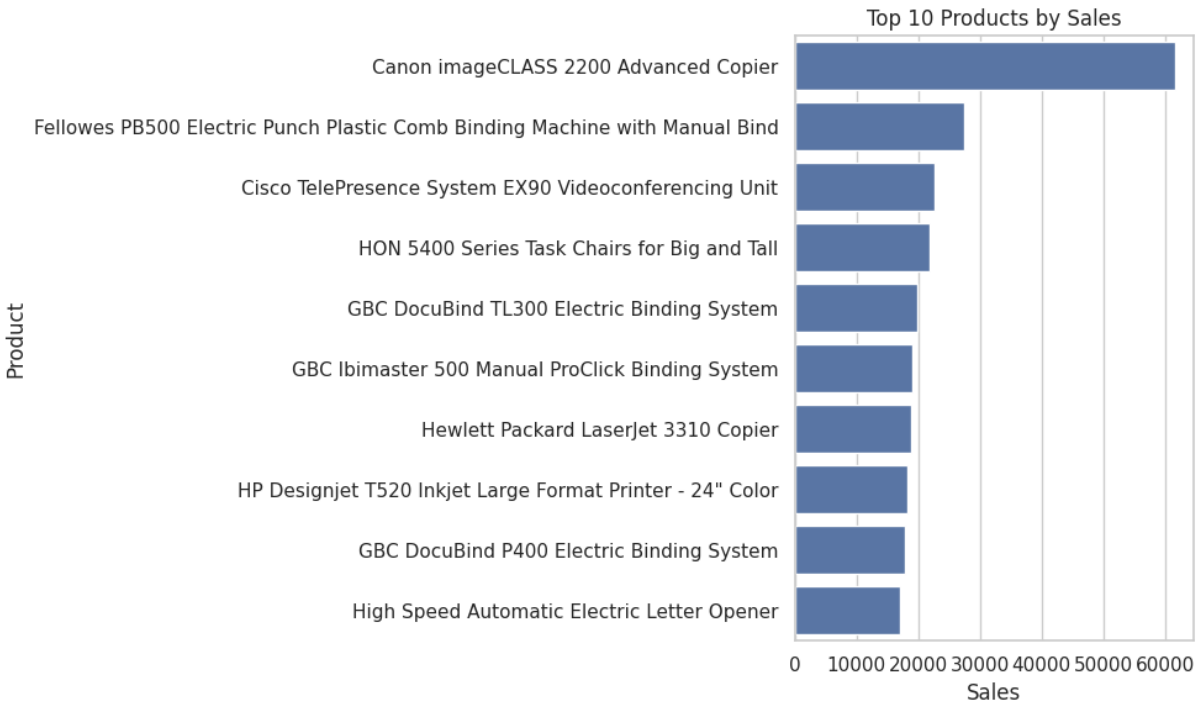
Code:

```
top_products = (
    df.groupby("Product Name")["Sales"]
      .sum()
      .sort_values(ascending=False)
      .head(10)
)

plt.figure(figsize=(10, 6))
sns.barplot(x=top_products.values, y=top_products.index)
plt.title("Top 10 Products by Sales")
plt.xlabel("Sales")
plt.ylabel("Product")
plt.tight_layout()
plt.show()

display(top_products.reset_index(name="Sales"))
```

```
<Figure size 1000x600 with 1 Axes>
```



	Product Name	Sales
0	Canon imageCLASS 2200 Advanced Copier	61599.824
1	Fellowes PB500 Electric Punch Plastic Comb Bin...	27453.384
2	Cisco TelePresence System EX90 Videoconferenci...	22638.480
3	HON 5400 Series Task Chairs for Big and Tall	21870.576
4	GBC DocuBind TL300 Electric Binding System	19823.479
5	GBC Ibimaster 500 Manual ProClick Binding System	19024.500
6	Hewlett Packard LaserJet 3310 Copier	18839.686
7	HP Designjet T520 Inkjet Large Format Printer ...	18374.895
8	GBC DocuBind P400 Electric Binding System	17965.068
9	High Speed Automatic Electric Letter Opener	17030.312

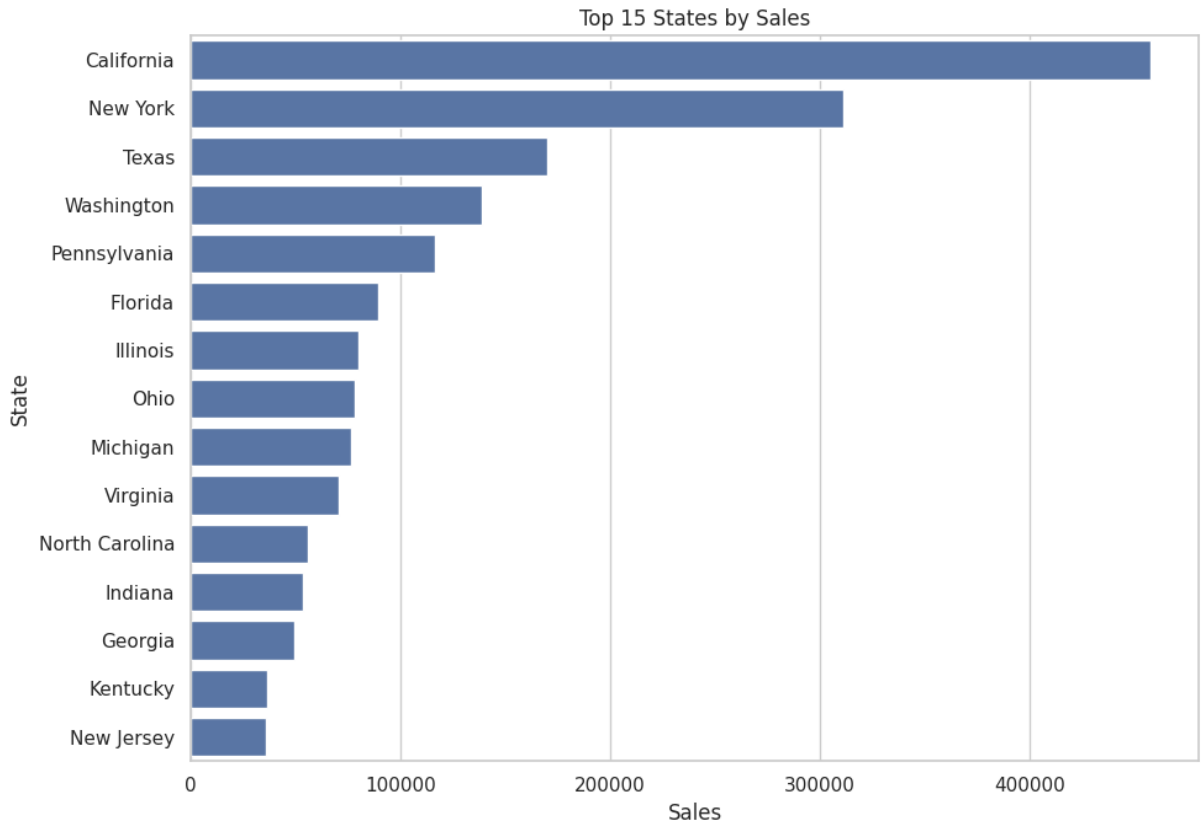
12. Sales by State

Code:

```
state_sales = (
    df.groupby("State")["Sales"]
      .sum()
      .sort_values(ascending=False)
      .head(15)
)

plt.figure(figsize=(10, 7))
sns.barplot(x=state_sales.values, y=state_sales.index)
plt.title("Top 15 States by Sales")
plt.xlabel("Sales")
plt.ylabel("State")
plt.tight_layout()
plt.show()
```

```
<Figure size 1000x700 with 1 Axes>
```



13. Interactive Filter Example

Code:

```
# Change these values to explore the dashboard data
selected_region = "West"
selected_category = "Technology"

filtered_df = df[
    (df["Region"] == selected_region) &
    (df["Category"] == selected_category)
]

print("Selected Region:", selected_region)
print("Selected Category:", selected_category)
print("Filtered rows:", len(filtered_df))

filtered_summary = pd.DataFrame({
    "Metric": ["Sales", "Profit", "Orders", "Customers"],
    "Value": [
        filtered_df["Sales"].sum(),
        filtered_df["Profit"].sum(),
        filtered_df["Order ID"].nunique(),
        filtered_df["Customer ID"].nunique()
    ]
})

display(filtered_summary)
```

```
      Metric        Value
0      Sales  251991.8320
1     Profit   44303.6496
2     Orders     490.0000
3  Customers     365.0000
```

14. Dashboard-Level Business Summary

Code:

```
region_profit = df.groupby("Region")["Profit"].sum()
category_profit = df.groupby("Category")["Profit"].sum()

best_sales_region = region_sales.idxmax()
best_profit_region = region_profit.idxmax()
best_sales_category = category_summary["Sales"].idxmax()
best_profit_category = category_summary["Profit"].idxmax()

print("BUSINESS INSIGHTS")
print("-----------------")
print(f"1. Highest-sales region: {best_sales_region}")
```

```
print(f"2. Highest-profit region: {best_profit_region}")
print(f"3. Highest-sales category: {best_sales_category}")
print(f"4. Highest-profit category: {best_profit_category}")
print(f"5. Total sales: ${total_sales:,.2f}")
print(f"6. Total profit: ${total_profit:,.2f}")
print(f"7. Number of orders: {number_of_orders:,}")
print(f"8. Number of customers: {customer_count:,}")
print(f"9. Average order value: ${average_order_value:,.2f}")
print(f"10. Profit margin: {profit_margin:.2f}%")
```

15. Dashboard Design Components

The dashboard should contain the following components when recreated in Tableau or Microsoft Power BI:

1. KPI cards for Total Sales, Total Profit, Orders, Customers, Average Order Value, and Profit Margin.

2. Sales by Region bar chart.

3. Sales and Profit by Category chart.

4. Monthly Sales Trend line chart.

5. Sales by Customer Segment chart.

6. Top Products chart.

7. Region and Category filters/slicers.

8. Drill-down from Region to State and from Category to Sub-Category/Product.

9. A clean dashboard layout with suitable titles and labels.

RESULT

An interactive business dashboard analysis was successfully prepared using the Superstore Sales Dataset. KPIs, sales and profit visualizations, time-based trends, customer segmentation, top products, filters, and business insights were generated. The prepared metrics and visualizations can be recreated as an interactive dashboard in Tableau or Microsoft Power BI.

Answers to Questions

Q1. What is Business Intelligence (BI)? How does it support organizational decision-making?

Business Intelligence is the process of collecting, analyzing, and presenting business data to support informed decisions. BI tools convert raw data into reports, dashboards, KPIs, and insights that help organizations monitor performance and identify trends.

Q2. What is an interactive dashboard? How is it different from a static report?

An interactive dashboard allows users to filter, explore, and drill down into data dynamically. A static report mainly presents fixed information and does not provide the same level of user interaction.

Q3. Compare Tableau and Microsoft Power BI based on features, usability, and business applications.

Tableau is widely used for interactive visual analytics and provides strong visualization and dashboard capabilities. Microsoft Power BI provides data modeling, interactive reporting, dashboards, and strong integration with Microsoft products. Both can connect to multiple data sources and support business analytics.

Q4. What are Key Performance Indicators (KPIs)? Give examples of KPIs used in sales analytics.

KPIs are measurable values used to evaluate business performance against objectives. Examples in sales analytics include Total Sales, Total Profit, Number of Orders, Customer Count, Average Order Value, and Profit Margin.

Q5. Explain the purpose of filters, slicers, and drill-down functionality in dashboards.

Filters and slicers allow users to display only the required subset of data. Drill-down allows users to move from summarized information to more detailed levels, such as Region to State to City or Category to Sub-Category.

Q6. What factors should be considered while designing an effective business dashboard?

Important factors include the purpose of the dashboard, target audience, relevant KPIs, data quality, suitable visualization types, clear layout, readability, consistent formatting, appropriate filters, and avoiding unnecessary visual complexity.

Q7. How can dashboards help managers monitor organizational performance in real time?

Dashboards can connect to regularly updated data sources and display current KPIs and visual trends. Managers can monitor sales, profit, orders, customers, and other metrics and quickly identify changes requiring attention.

Q8. Why is data visualization important in Business Intelligence tools?

Data visualization presents complex information in graphical form. It makes trends, comparisons, relationships, and outliers easier to understand and helps business stakeholders interpret analytical results quickly.

Q9. What insights can be derived from a sales dashboard for an e-commerce company?

A sales dashboard can show total revenue, profit, order volume, customer performance, regional performance, top products, product categories, sales trends, and areas with low performance. These insights can support inventory, marketing, sales, and planning decisions.

Q10. Explain how interactive dashboards improve communication between data analysts and business stakeholders.

Interactive dashboards provide a common visual representation of business information. Stakeholders can apply filters, explore details, and focus on the metrics relevant to them. This makes analytical findings easier to understand and discuss.